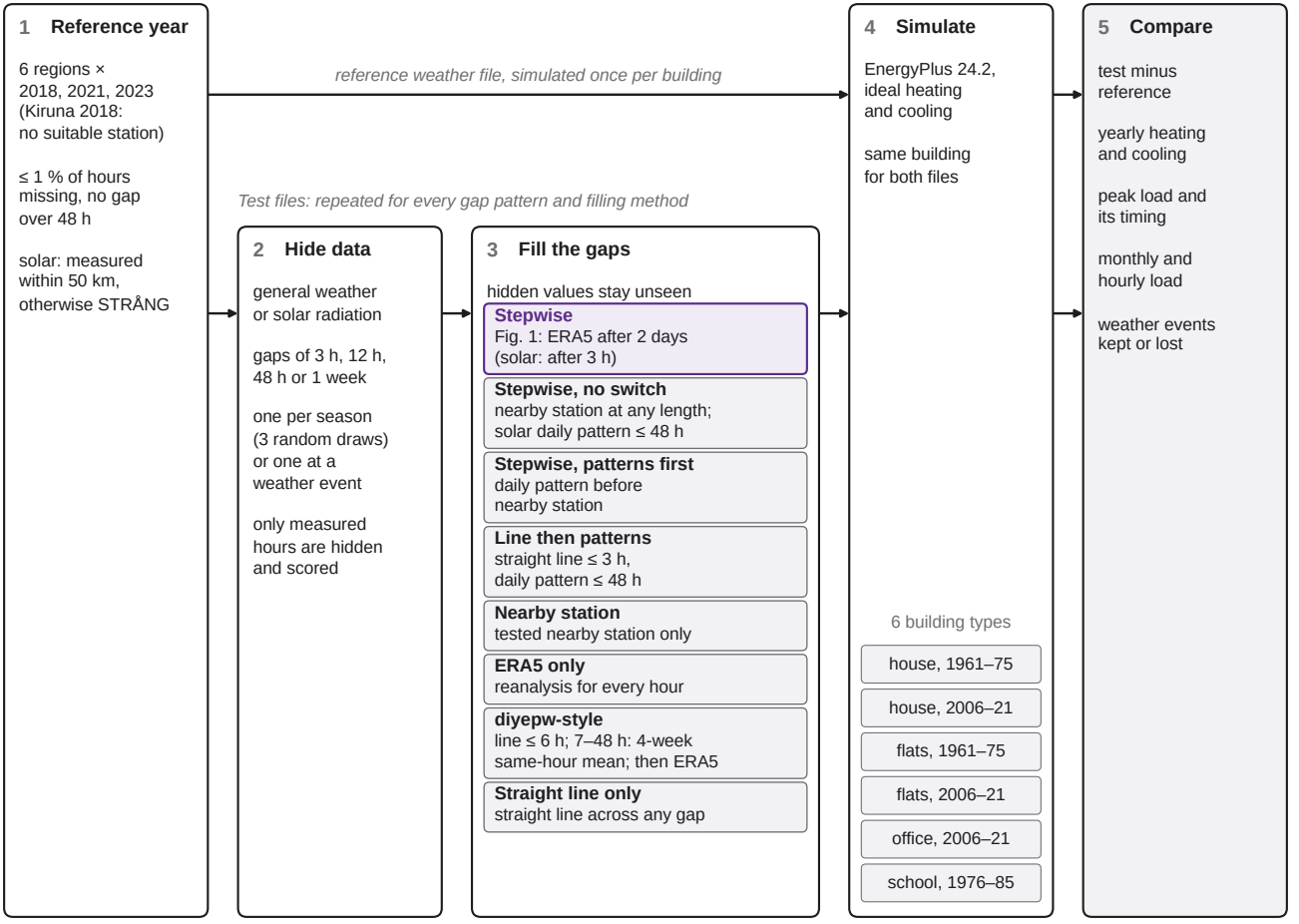
1 Reference year
6 regions × 2018, 2021, 2023 (Kiruna 2018: no suitable station)
≤ 1 % of hours missing, no gap over 48 h
solar: measured within 50 km, otherwise STRÅNG
reference weather file, simulated once per building
Test files: repeated for every gap pattern and filling method
2 Hide data
general weather or solar radiation
gaps of 3 h, 12 h, 48 h or 1 week
one per season (3 random draws) or one at a weather event
only measured hours are hidden and scored
3 Fill the gaps
hidden values stay unseen
Stepwise
Fig. 1: ERA5 after 2 days (solar: after 3 h)
Stepwise, no switch
nearby station at any length; solar daily pattern ≤ 48 h
Stepwise, patterns first
daily pattern before nearby station
Line then patterns
straight line ≤ 3 h, daily pattern ≤ 48 h
Nearby station
tested nearby station only
ERA5 only
reanalysis for every hour
diyepw-style
line ≤ 6 h; 7–48 h: 4-week same-hour mean; then ERA5
Straight line only
straight line across any gap
4 Simulate
EnergyPlus 24.2, ideal heating and cooling
same building for both files
6 building types
house, 1961–75
house, 2006–21
flats, 1961–75
flats, 2006–21
office, 2006–21
school, 1976–85
5 Compare
test minus reference
yearly heating and cooling
peak load and its timing
monthly and hourly load
weather events kept or lost